Subject: RE: [EXTERNAL] *Reminder* FPP run group meeting
Date: Thursday, October 22, 2026 at 1:29:32 PM Eastern Daylight Time
From: Andrew Puckett
To: Ed Brash, Garth Huber, Steffen Strauch
Attachments: image.png, image.png

Thanks, Ed! To my mind, the major potential show stoppers are going to be (a) the trigger rates, (b) the background rate/pileup situation in the NPS calorimeter, and (c) perhaps also the effect of the high background rate on the HMS tracking under such high-rate conditions. As in GEp-III, parts of the standard HMS detector package would need to come out to accommodate the FPP, so I wonder about the singles rates in HMS. I'd like to avoid repeating the GEp-III mistake of sticking a scintillator paddle upstream of the HMS drift chambers, killing our angular resolution. The HMS "S1" scintillator planes would likely face pretty high counting rates at the proposed luminosity, so a pure S1 trigger in HMS in coincidence with an NPS ECAL trigger might not be selective enough. Perhaps we can find a way to fit the FPP in while keeping the S2 plane in the HMS detector stack. This might be easier if we don't need both CH2 analyzers. Perhaps locating them behind the FPP drift chambers is an option. Another option is including the HMS calorimeter in the trigger, but as in GEp-III, we should prefer to avoid that, to avoid introducing trigger bias in the FPP angular distribution. I also have to imagine the calorimeter rates being quite high at an angle of 27-ish degrees at a distance of 3 meters from the target at that luminosity (the SBS ECAL was at the same central angle at 6 meters from the target, but the size of its blocks was about twice as big as the NPS blocks, so similar solid angle per block. Rad-hardness is one thing, but background counting rates and pileups could complicate the energy resolution, trigger, and offline analysis. The rad-hardness of the NPS calorimeter and all the fancy modern digital triggering capability we have now should help, and if the NPS energy resolution is sufficiently good, we should be able to use a pretty high threshold on the electron (calorimeter) trigger, presumably better than we were able to achieve in practice in SBS. At the same time, I have to imagine that the Hall C DAQ rate capability is much higher by now, assuming that all the old fastbus electronics have (hopefully) been replaced?

In my mind, the major work toward a full proposal development consists of:

(a) optimizing the choice of kinematics for Q^2 reach and precision goals, under the assumption that the HMS/NPS combination can handle something like 70-75 uA on a 30-cm LH2 target.
(b) Estimating the background and trigger rates in the detectors, with a focus on the trigger detectors and the HMS drift chambers, hopefully by interpolating from operational experience in the 12 GeV era with both the HMS and the NPS, and coming up with a viable/plausible plan for the trigger and DAQ.
(c) We may also wish to evaluate the FPP-in-SHMS option and even higher Q^2 values,

but I suspect that both are going to be considerably less attractive than the HMS option given the smaller solid angle acceptance of the SHMS.

Best regards,
Andrew

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From: Ed Brash <brash@jlab.org>
Date: Thursday, October 22, 2026 at 12:39 PM
To: Andrew Puckett <puckett@jlab.org>; Garth Huber <huberg@uregina.ca>; Steffen Strauch <strauch@sc.edu>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

Hi Andrew et al.,

This is excellent! Thanks for working this up so quickly!

At the very least, I consider this justification to continue with more detailed studies!

Following up on Dave Gaskell's suggestion, inviting Mark Jones to the next meeting is the best idea, and I will, before that meeting, catch up with him and get a sense of the status/existence of the Hall C FPP components.

Cheers,
E.

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From: Andrew Puckett <puckett@jlab.org>
Date: Wednesday, October 21, 2026 at 7:10 PM
To: Garth Huber <huberg@uregina.ca>; Ed Brash <brash@jlab.org>; Steffen Strauch

<strauch@sc.edu>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

PS—At Q^2 = 11, the spin precession angle in the HMS will be about 325 degrees, which is suboptimal but still gives a sin(chi) of 0.58. And as we know from the GEp-III experience, in a focusing spectrometer like HMS, even if we had 360-degree central precession, the precision on the FF ratio would still be dominated by the measurement of P_T, since P_L is so large and chi acceptance of HMS is so large. This is in contrast to the SBS, where the precession angle was about 75 degrees, and roughly constant within the acceptance, because it's a non-focusing dipole spectrometer. In the HMS, the precession angle will vary quite a lot within the acceptance due to its focusing optics with point-to-point tune. So precession doesn't alter the conclusion that GEP-4 may, in fact, kick GEP-5's ass in the end.

Cheers,
Andrew

From: Andrew Puckett <puckett@jlab.org>
Date: Wednesday, October 21, 2026 at 6:55 PM
To: Garth Huber <huberg@uregina.ca>; Ed Brash <brash@jlab.org>; Steffen Strauch <strauch@sc.edu>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

Just a quick follow-up from our earlier discussion; I downloaded the latest SIMC from GitHub, and ran a 10k-event H(e,e'p) simulation for the kinematics of the attached screenshot. These are almost identical to the SBS kinematics. It assumes the NPS calorimeter at 3 meters from the target at an angle 26.7 degrees (I don't know how realistic that is). At that distance, the NPS calorimeter APPEARS to more or less comfortably cover the required solid angle; it could probably even be pushed back a little bit. I'm not 100% sure if I'm interpreting the calorimeter acceptance in SIMC correctly, but the attached screenshot does appear to show that the envelope of electrons on the NPS calorimeter at a distance of 3 m covers roughly +/- 30 cm vertically and +/-25 cm horizontally. Whether it could handle the background rate and radiation environment of 75 uA on 30-cm LH2 is another question. But I suspect the angular resolution would be good enough at that distance (the blocks are ~2 x 2 cm cross section, assume spatial resolution ~5 mm / 3 m gives 1.6 mrad angular resolution, which seems pretty comparable to the kind of angular resolution we had in SBS, and of course in the HMS we will have ~10-20X better momentum resolution for the scattered proton.

I'd say this looks really promising.

In 45 PAC days at 75 uA on 30-cm LH2 we would collect about 12M elastic ep events. Note that the "best case scenario" for the SBS measurement is based on the assumption that

we recover something like ~9.3M events in the final analysis.

The polarimeter FOM should be quite similar (probably better, in fact), as the kinematics are virtually identical to the SBS measurement, the CH2 thickness is about the same, and the polar and azimuthal acceptance (and uniformity) of the Hall C FPP for the secondary scattering is likely to be considerably greater in practice than that of the SBS polarimeter.

So indeed, with 45 PAC days with HMS+FPP+NPS calorimeter, we can do about as well as (in fact, slightly better) than the BEST-CASE scenario for what the SBS actually achieved at 11 GeV^2. And it's likely that the SBS measurement will actually prove to have fallen short of its "best-case" outcome. And the SBS measurement at Q^2 = 11 actually received the equivalent of 53 PAC days (in terms of floor time; and that was some pretty high-efficiency running for CEBAF).

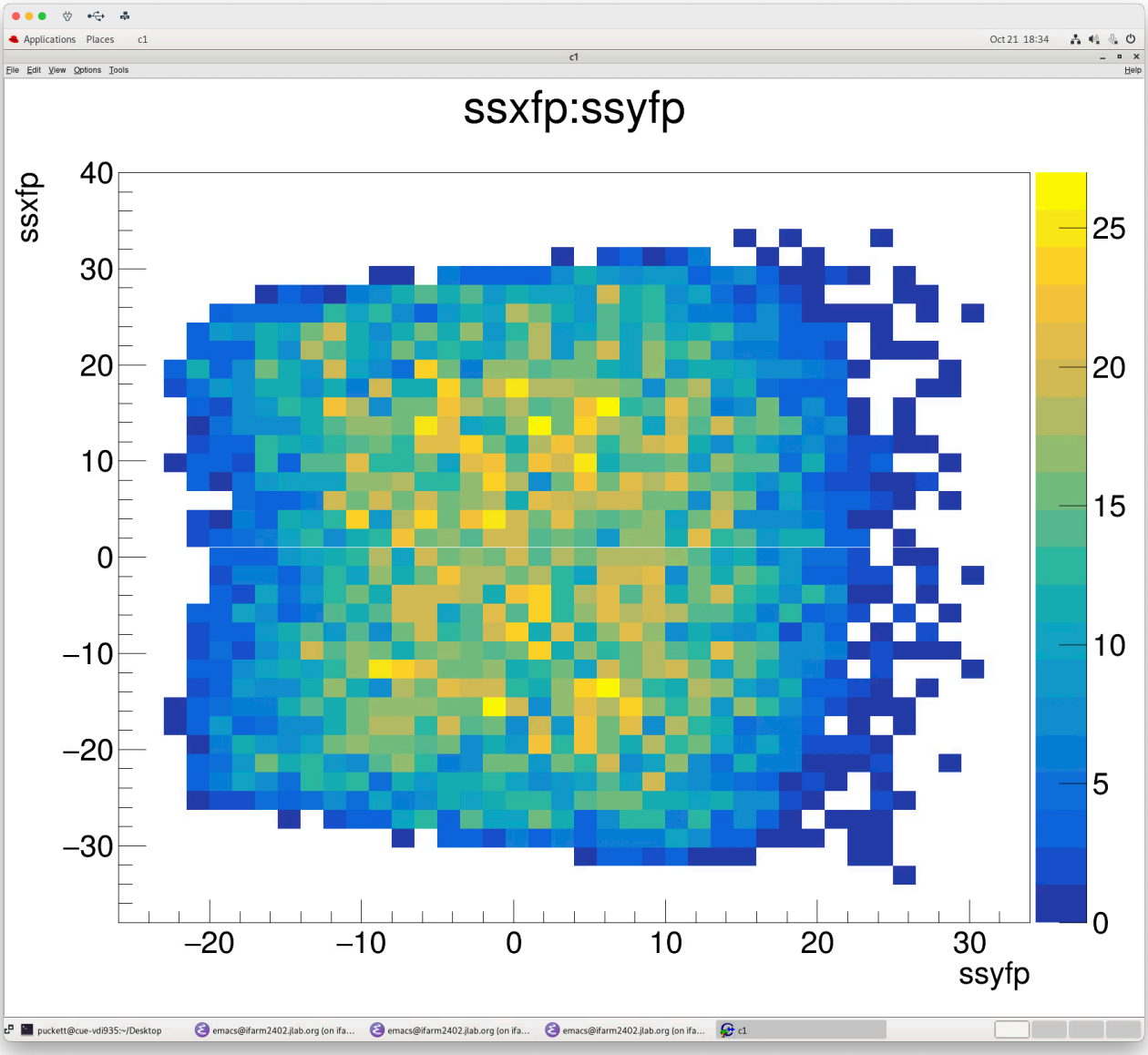
So I'm starting to believe we might be able to sell a GEP4 reboot to the PAC, especially if, by the time of the next PAC, we have a better read on what the SBS measurement actually achieved. By that time, we might be looking at a scenario where we can do something like a factor of 2 better than the SBS measurement at the same Q^2.

That's ASSUMING that the NPS calorimeter would work well in these kinematics…

These numbers are VERY tentative and very "quick-and-dirty" at the moment, so please keep them within this core group for the moment. I also don't want to create more political problems for future SBS running and/or other concepts like SoLID by showing that after more than 15 years of investment in building and running SBS with GEP as the "flagship" measurement, that it turns out that the PAC backed the wrong horse and should have backed GEP-4 to start with. At the time, of course, the NPS calorimeter wasn't a thing.

Cheers,
Andrew

	Mp	0.93827															
	Ebeam	Q2	nu	Eprime	ethetadeg	pthetadeg	pp	epsilon	tau	F2/F1 (fro	GE/GM	PT	PL	SIMC rate	PAC days	events	
	10.7	11	5.86184	4.83816	26.6507	18.7971	6.73507	0.68367	3.12374	0.32051	-0.0009	0.00034	0.72979	3.25	45	12636000	
	8.6	8	4.26316	4.33684	26.7788	22.4528	5.1161	0.72951	2.27181	0.40027	0.06475	-0.02695	0.68305				
Commissioning?																	



From: Garth Huber <huberg@uregina.ca>
Date: Tuesday, October 20, 2026 at 12:23 PM
To: Ed Brash <brash@jlab.org>; Steffen Strauch <strauch@sc.edu>; Andrew Puckett <puckett@jlab.org>
Subject: [EXTERNAL] *Reminder* FPP run group meeting

This is a reminder of our meeting on Wednesday at 15:00 Eastern.

Garth Huber is inviting you to a scheduled Zoom meeting.

Topic: FPP Run Group Meeting
Time: Oct 21, 2026 13:00 Saskatchewan

Join Zoom Meeting
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